OLS Assumptions at a Glance: SLR.1 – 5, MLR.1 – 6

SLR has one explanatory variable; MLR has several. What each assumption buys, in finite and large samples.

#	Simple (SLR)	Multiple (MLR)	Role / exam point
1	Linear in parameters	Linear in parameters	**Linear in coefficients**; regressors may be logged or squared.
2	Random sampling	Random sampling	Observations are **randomly sampled** from the population.
3	Sample variation in x $\Sigma(x_i-\bar{x})^2 > 0$	No perfect collinearity No constant or exact linear dependence	SLR: x must vary. MLR: **exact collinearity prevents identification.**
4	Zero conditional mean $E(u\|x)=0$	Zero conditional mean $E(u\|x_1,\cdots,x_k)=0$	**Core of unbiasedness.** Consistency needs only the weaker **MLR.4'**: $E(u)=0$, $Cov(x_j,u)=0$ (Ch. 5).
5	Homoskedasticity $Var(u\|x)=\sigma^2$	Homoskedasticity $Var(u\|X)=\sigma^2$	Needed for **classical variances, BLUE,** and **large-sample t/F** (Ch. 5).
6	—	Normality of errors $u\|X \sim Normal(0,\sigma^2)$	**Exact** t/F in any sample size; **not needed in large samples** (Ch. 5).

1 – 4 → unbiased and consistent; 1 – 5 → BLUE and large-sample t/F; add MLR.6 → exact t/F in any sample.

Sources: Chapter 2, slides 28 – 36; Chapter 3, slides 18 – 22, 36 – 38; Chapter 4, slides 5 – 12; Chapter 5, slides 4 – 10.

Bias and Variance: Simple vs. Multiple Regression OLS

OLS is an estimation method; SLR and MLR describe the model. Theorems 2.2 and 3.2 give the slope variances.

Comparison	Simple regression (SLR)	Multiple regression (MLR)
Source of bias	$E(u\|x) \neq 0$. If x is related to factors left in u (endogeneity), the slope is biased.	$E(u\|x_1,\cdots,x_k) \neq 0$. An omitted variable that affects y and is correlated with an included x causes **omitted-variable bias.**
Example	Wage on education: **omitted ability** is related to both, so the education slope is biased.	Controls may reduce bias, but **remaining omitted factors or a wrong functional form** can still violate MLR.4.
Slope variance	$\mathbf{Var(\hat{\beta}_1\|x) = \sigma^2 / \Sigma(x_i-\bar{x})^2}$ Theorem 2.2 · SLR.1 – SLR.5	$\mathbf{Var(\hat{\beta}_j\|X) = \sigma^2 / [SST_j(1-R_j^2)]}$ Theorem 3.2 · MLR.1 – MLR.5 · R_j^2: R^2 of x_j on the other x's
Variance drivers	$\sigma^2 \uparrow$ → **variance ↑**. More variation in x → variance ↓. No multicollinearity term.	$\sigma^2 \uparrow$ → **variance ↑**. $SST_j \uparrow$ → variance ↓. $R_j^2 \uparrow$ → **variance ↑** (multicollinearity).
Exam trap	Heteroskedasticity breaks the variance formula and standard errors, but **does not cause bias if E(u|x)=0 holds.**	Unless perfect, multicollinearity **raises variance, not bias.** Perfect collinearity **violates MLR.3: no identification.**

Simple regression is the $R_j^2 = 0$ case of Theorem 3.2. Omitted factors correlated with x → **bias**; large σ^2, little x variation, strong collinearity → **variance.**

Sources: Chapter 2, slides 31 – 36; Chapter 3, slides 18 – 29, 31 – 33.

Simple Regression: Model, OLS, and Fit

What β_1 measures, when it is causal, and what OLS guarantees in every sample

Model and causal reading

$y = \beta_0 + \beta_1 x + u$

u: all unobserved factors. If $\Delta u = 0$, then $\Delta y = \beta_1 \Delta x$ (ceteris paribus).

$E(u|x) = E(u) = 0$ (conditional mean independence)

If ability sits in u and is related to education, this fails → β_1 is not a causal effect. A fitted line alone never establishes causality.

Population regression function

$E(y|x) = \beta_0 + \beta_1 x$

A one-unit increase in x changes the mean of y by β_1.

OLS: minimize $\Sigma \hat{u}_i^2$

$\hat{\beta}_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

Same sign as the sample correlation; requires $\Sigma(x_i - \bar{x})^2 > 0$.

Algebraic properties (any sample)

$\Sigma \hat{u}_i = 0$, $\Sigma x_i \hat{u}_i = 0$, and $(\bar{x}, \bar{y})$ lies on the fitted line.

Goodness of fit

$SST = SSE + SSR$, $R^2 = SSE/SST = 1 - SSR/SST$

A low R^2 does not make causal estimates useless; a high R^2 does not establish causality.

Log forms change how β_1 is read — see the table in Chapter 6.

A causal reading of β_1 rests on $E(u|x)=0$; OLS minimizes the sum of squared residuals and its algebraic properties need no assumptions.

Sources: Chapter 2, slides 5 – 26 · Supplement: OLS formulas (standard textbook results)

Unbiasedness and Variance (SLR.1 – 5)

β is a random variable that changes with the sample: is it right on average, and how much does it vary?

Theorem 2.1 — Unbiasedness (SLR.1 – 4)

$E(\hat{\beta}_0) = \beta_0, \quad E(\hat{\beta}_1) = \beta_1$

Right on average across repeated samples. Key: SLR.4, E(u|x)=0.

Homoskedasticity (SLR.5)

$Var(u|x) = \sigma^2$

Counter-example: wage variance rising with education. Breaks the variance formula, not unbiasedness.

FROM LECTURE · ECONOMETRICS 1 — POPULATION AND SAMPLE

Population: the whole set we want to study (may be hypothetical).
Sample: the subset we analyze.

The population is often too costly to study, so the sample tells us **what the population is like.**

Is the sample **representative? External validity:** do conclusions hold outside the study's context?

Theorem 2.2 — Variances of OLS (SLR.1 – 5)

$Var(\hat{\beta}_1|x) = \sigma^2 / SST_x, \quad SST_x = \Sigma(x_i - \bar{x})^2$

$Var(\hat{\beta}_0|x) = \sigma^2 \cdot n^{-1}\Sigma x_i^2 / SST_x$

Higher σ^2 → higher variance; more variation in x → lower variance. A larger sample raises SST_x.

Theorem 2.3 — Error variance and standard errors

$\hat{\sigma}^2 = SSR / (n - 2), \quad E(\hat{\sigma}^2) = \sigma^2$

$se(\hat{\beta}_1) = \hat{\sigma} / \sqrt{SST_x}$

n − 2: degrees of freedom lost by estimating β_0 and β_1. $\hat{\sigma}$ is the standard error of the regression (SER).

u is unobserved, so σ^2 is estimated from residuals — se is an estimate of sd.

SLR.2 (random sampling) is what lets the sample speak for the population.

Unbiasedness needs SLR.1 – 4; the variance formula also needs SLR.5. Precision rises with small σ^2 and large variation in x.

Sources: Chapter 2, slides 27 – 38; Econometrics 1, slides 1 – 4 · Supplement: Var(β_0) formula

Binary Regressors, Potential Outcomes, Random Assignment

With a 0/1 regressor, regression compares group means; with random assignment, it estimates a treatment effect

Binary (dummy) explanatory variable

$y = \beta_0 + \beta_1 x + u, \quad x \in \{0, 1\}$

$\beta_0 = E(y|x=0), \quad \beta_1 = E(y|x=1) - E(y|x=0)$

OLS: $\hat{\beta}_0$ = control-group mean; **$\hat{\beta}_1$ = difference in group means.** OLS properties are unchanged when x is binary.

Potential outcomes

$te_i = y_i(1) - y_i(0), \quad \tau = E[y_i(1) - y_i(0)]$ (ATE)

For any unit we observe $y_i(1)$ or $y_i(0)$, **never both** → we estimate the average treatment effect (ATE).

Observed: $y_i = (1 - x_i)y_i(0) + x_i y_i(1)$

Link to regression

With a constant effect τ, $y_i = y_i(0) + \tau x_i$ → **regressing y on x estimates τ.**

Under **random assignment,** x is independent of $y_i(0)$ → OLS is an **unbiased estimator of τ.**

Random assignment (RCTs): meaning and limits

Assign units so there are **no systematic differences** between groups other than treatment. RCTs are costly and may raise ethical issues, but **imagining the ideal experiment** sharpens the causal question.

Example: job training and earnings

Participants earn **$1,790 more,** a **39.3% increase** over the $4,550 average of non-participants.

A dummy slope is a **difference in group means;** it is an average treatment effect **only under random assignment.**

Sources: Chapter 2, slides 39 – 43 · Supplement: observed-outcome identity $y=(1-x)y(0)+xy(1)$

Multiple Regression: Partial Effects and Partialling Out

The coefficient on x_1 is a partial slope — generally not the slope of y on x_1 alone

Model and interpretation

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + u$

$\Delta\hat{y} = \hat{\beta}_1 \Delta x_1 \quad (\Delta x_2 = \cdots = \Delta x_k = 0)$

Regression isolates the effect **holding other x's fixed** — but **not** the unobservables in u. Same ACT, 1 point higher hsGPA → colGPA **0.453 higher.**

Partialling out

Regress x_j on the other x's → residuals $\hat{r}_{ij}$; then

$\hat{\beta}_j = \Sigma \hat{r}_{ij} y_i \,/\, \Sigma \hat{r}_{ij}^2$

$\hat{\beta}_j$ uses only the **unique variation** in x_j.

R^2 in multiple regression

Adding a regressor never lowers R^2; a small R^2 can still give good ceteris paribus estimates.

FROM LECTURE · ECONOMETRICS 2 — SLOPE AND PARTIAL SLOPE

Regressing y on x_1 alone: **β_1 is the slope.**

In a multiple regression, **β_1 might not be the slope — β_1 is the partial slope.**

When the other regressor moves with x_1, **β_1 is not the slope** of the simple y–x_1 picture.

Simple slope vs. partial slope

$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_2 \tilde{\delta}_1 \quad (\tilde{\delta}_1$: slope of x_2 on $x_1)$

Equal only if $\hat{\beta}_2 = 0$ or $\tilde{\delta}_1 = 0$. The simple slope also carries x_2's effect through the correlation — the in-sample version of OVB.

$\hat{\beta}_j$ is a **partial slope** from the unique variation in x_j; the simple slope differs by **$\hat{\beta}_2 \tilde{\delta}_1$.**

Sources: Chapter 3, slides 5 – 17; Econometrics 2, slides 1 – 6 · Supplement: $\beta_1 = \beta_1 + \beta_2 \delta_1$ (standard textbook result)

MLR Assumptions, Unbiasedness, Omitted-Variable Bias

Which variables go in or out determines bias

Theorem 3.1 — Unbiasedness (MLR.1 – 4)

- MLR.1 linear in parameters · MLR.2 random sampling
- MLR.3 no perfect collinearity: no constant regressor, **no exact linear relation** (e.g., $x_3 = x_1 + x_2$); needs $n \geq k+1$.
- MLR.4 $E(u|x_1,\cdots,x_k) = 0$: **more plausible than in simple regression.**

$E(\hat{\beta}_j) = \beta_j, \quad j = 0, 1, \cdots, k$

MLR.3 rules out only **perfect** collinearity — high correlation is allowed.

Including an irrelevant variable

Keeps OLS unbiased; the cost is **possibly higher variance.**

General case

With several regressors, omitting one **generally biases all coefficients**; the direction is hard to sign.

Omitted-variable bias (OVB) — simple case

True: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + u$ Estimated: y on x_1 only

$E(\tilde{\beta}_1) = \beta_1 + \beta_2\tilde{\delta}_1$ → bias = $\boldsymbol{\beta_2 \cdot \tilde{\delta}_1}$

No bias if $\beta_2 = 0$ or x_1 and x_2 are uncorrelated.

	$Corr(x_1, x_2) > 0$	$Corr(x_1, x_2) < 0$
$\boldsymbol{\beta_2 > 0}$	Positive bias	Negative bias
$\boldsymbol{\beta_2 < 0}$	Negative bias	Positive bias

Omitting ability in a wage equation: ability raises wages (+) and correlates with education (+) → **upward bias in the return to education.**

An omitted variable that **affects y and correlates with an included x** causes bias, signed by β_2 and the correlation.

Sources: Chapter 3, slides 18 – 26 · Supplement: OVB formula and sign table (standard textbook result)

OLS Variance, Multicollinearity, Gauss – Markov

Three drivers of precision, the bias – variance tradeoff, and why OLS is BLUE

Theorem 3.2 — Slope variances (MLR.1 – 5)

$\mathrm{Var}(\hat{\beta}_j|X) = \sigma^2 / [SST_j(1 - R_j^2)]$

- σ^2: more noise → higher variance.
- SST_j: more variation in x_j → lower variance; grows with n.
- R_j^2: others explain x_j better → higher variance.

Multicollinearity

$VIF_j = 1 / (1 - R_j^2)$

Precision, not bias; not an MLR.3 violation. Only the variables involved are affected.

Bias – variance tradeoff

$\beta_2 = 0$: dropping x_2 lowers $\mathrm{Var}(\tilde{\beta}_1)$. $\beta_2 \neq 0$: dropping x_2 biases $\tilde{\beta}_1$ — **the bias remains as n grows.**

Theorem 3.3 — Error variance and se

$\hat{\sigma}^2 = SSR / (n - k - 1)$

$\mathrm{se}(\hat{\beta}_j) = \hat{\sigma} / \sqrt{[SST_j(1 - R_j^2)]}$

$n - k - 1$ = **degrees of freedom.** This se is **valid only under homoskedasticity (MLR.5).**

Theorem 3.4 — Gauss – Markov

Under MLR.1 – 5, OLS is **BLUE**: linear ($\Sigma w_{ij} y_i$), unbiased, and **smallest variance** among linear unbiased estimators.

Under heteroskedasticity, OLS stays unbiased but is no longer best.

Controls and causal questions

Treatment w plus controls x approximates **random assignment conditional on observables.**

Collinearity inflates variance; omitted variables cause bias. Under MLR.1 – 5 OLS is **BLUE.**

Sources: Chapter 3, slides 27 – 41 · Supplement: VIF formula

Normal Errors, the t Distribution, and t Tests

MLR.6 gives exact sampling distributions; choose the direction of H_1 before seeing results

Alternative H_1 (H_0: $\beta_j = 0$)	Type	Rejection rule (level α, c: critical value)	Slide example (5%)
$\beta_j > 0$	Right-tailed	$t > c$ (upper tail α)	df = 28: $t > 1.701$
$\beta_j < 0$	Left-tailed	$t < -c$ (lower tail α)	df = 18: $t < -1.734$
$\beta_j \neq 0$	Two-tailed	$\lvert t \rvert > c$ ($\alpha/2$ in each tail)	$\lvert t \rvert > 2.06$

MLR.6 and the CLM

$u \sim \text{Normal}(0, \sigma^2)$, independent of the x's. MLR.1 – 6 = **classical linear model.** Then $\hat{\beta}_j$ is normal (Thm 4.1) and

$(\hat{\beta}_j - \beta_j) / se(\hat{\beta}_j) \sim t(n - k - 1)$ (Thm 4.2)

Normality fails for wages, counts, 0/1 outcomes — **logs** can help; Ch. 5 drops MLR.6.

General hypothesis H_0: $\beta_j = a_j$

$t = (\hat{\beta}_j - a_j) / se(\hat{\beta}_j)$

p-values

The smallest significance level at which H_0 is rejected. Two-sided: $p = P(\lvert T \rvert > \lvert t \rvert)$.

$P(\lvert T \rvert > 1.85) = 2 \times 0.0359 = \mathbf{0.0718} \rightarrow$ reject at 10%, not at 5%.

$p < \alpha \Leftrightarrow$ reject H_0. Say "fail to reject," **never "accept H_0."** Large df, two-sided 10/5/1%: 1.645 / 1.96 / 2.576.

School size: insignificant in levels → **significant in logs**; functional form matters.

Under MLR.1 – 6, $t = (\hat{\beta}_j - a_j)/se \sim t(n-k-1)$; reject with the region that matches H_1, or when $p < \alpha$.

Sources: Chapter 4, slides 5 – 27 · Supplement: large-sample critical values (1.645/1.96/2.576)

Confidence Intervals, Economic Significance, Linear Combinations

Intervals show precision; reparameterization gives the standard error of a coefficient difference directly

Confidence intervals

$\hat{\beta}_j \pm c \cdot se(\hat{\beta}_j)$

c: 97.5th percentile of t(n−k−1); large df: 1.96 (90%: 1.645; 99%: 2.576).

The bounds are **random.** In repeated samples, **95% of intervals built this way cover the true β_j.**

a_j outside the 95% CI ⇔ **reject H_0: $\beta_j = a_j$** in a 5% two-sided test.

Statistical vs. economic significance

- If significant, judge **magnitude and units.**
- **Significant ≠ important:** large n makes small effects significant.
- Important but **wrong sign** → suspect misspecification.

Testing a linear combination

$\log(wage) = \beta_0 + \beta_1 jc + \beta_2 univ + \beta_3 exper + u$

$H_0: \beta_1 = \beta_2$ vs $H_1: \beta_1 < \beta_2$, $t = (\hat{\beta}_1 - \hat{\beta}_2) / se(\hat{\beta}_1 - \hat{\beta}_2)$

$se(\hat{\beta}_1 - \hat{\beta}_2)$ needs **$Cov(\hat{\beta}_1, \hat{\beta}_2)$,** which is not in standard output.

Fix: reparameterize

$\theta_1 = \beta_1 - \beta_2 \rightarrow \beta_1 = \theta_1 + \beta_2$

$\log(wage) = \beta_0 + \theta_1 jc + \beta_2 (jc + univ) + \beta_3 exper + u$

Create totcoll = jc + univ: **the coefficient and se on jc are $\hat{\theta}_1$ and $se(\hat{\theta}_1)$.** Works for any single linear hypothesis.

A 95% CI covers the truth in **95% of repeated samples**; compare coefficients by **reparameterizing.**

Sources: Chapter 4, slides 28 – 35 · Supplement: large-sample critical values

F Tests: Multiple Linear Restrictions

Ask whether imposing the restrictions raises SSR by "enough"

Testing exclusion restrictions

H_0: $\beta_{k-q+1} = \cdots = \beta_k = 0$ (q restrictions)

Estimate the unrestricted model (ur) and the restricted model (r, q variables dropped).

$F = [(SSR_r - SSR_{ur})/q] / [SSR_{ur}/(n - k - 1)] \sim F(q, n - k - 1)$

Dropping variables can only **raise** SSR → $F \geq 0$. **Reject H_0 if $F > c$** (upper-α critical value), or if $p < \alpha$.

R^2 form (same dependent variable only):

$F = [(R^2_{ur} - R^2_r)/q] / [(1 - R^2_{ur})/(n - k - 1)]$

Individually insignificant, jointly significant

Highly correlated regressors can each have small t yet be **jointly significant** → use F, not repeated t tests.

Overall significance of the regression

H_0: $\beta_1 = \cdots = \beta_k = 0$, $F = (R^2/k) / [(1 - R^2)/(n - k - 1)]$

General linear restrictions

Rational house-price assessments: assessment coefficient = 1, others = 0. The restriction **changes the dependent variable → use only the SSR form.**

With one restriction (q = 1), **$F = t^2$** — same conclusion as the two-sided t test.

Test several restrictions with **F based on the rise in SSR**; if the dependent variable changes, use the **SSR form.**

Sources: Chapter 4, slides 36 – 43 · Supplement: $F = t^2$

Consistency and Large-Sample Inference

What OLS guarantees as n grows — without normal errors, but still with MLR.1 – 5

Consistency

$\text{plim}\ \hat{\beta}_j = \beta_j$

The probability of being **arbitrarily close to the truth** → 1 as n grows. A minimum requirement for any estimator.

Theorem 5.1 and MLR.4'

$\text{plim}\ \hat{\beta}_1 = \beta_1 + \text{Cov}(x_1, u) / \text{Var}(x_1)$

MLR.4': $E(u) = 0$, $\text{Cov}(x_j, u) = 0$

Zero **correlation** suffices → OLS may be biased but **consistent.**

Asymptotic omitted-variable bias

$\text{plim}\ \tilde{\beta}_1 = \beta_1 + \beta_2\delta_1, \quad \delta_1 = \text{Cov}(x_1, x_2) / \text{Var}(x_1)$

Inconsistency does not shrink with n.

Theorem 5.2 — Asymptotic normality (MLR.1 – 5)

$(\hat{\beta}_j - \beta_j) / \text{se}(\hat{\beta}_j) \ \sim^a\ \text{Normal}(0, 1)$

t tests, CIs, and F tests are valid in large samples without MLR.6. MLR.1 – 5 are still necessary — especially homoskedasticity.

Why large samples are better

$\text{se}(\hat{\beta}_j) \approx c_j / \sqrt{n}$

Quadrupling n roughly halves the se (birth-weight example).

LM statistic for q exclusion restrictions

$LM = n \cdot R^2_u \ \sim^a\ \chi^2(q)$

R^2_u: from regressing restricted-model residuals on **all** x's. Testing normality gives **no actionable response.**

MLR.1 – 3 + MLR.4' → consistency; MLR.1 – 5 → large-sample t/F without normality. OVB does not vanish as n grows.

Sources: Chapter 5, slides 2 – 12 · Supplement: plim formulas, MLR.4', se ≈ c/√n, LM statistic (equations are images in the slides)

Units, Standardized Coefficients, Logs

Rescaling changes coefficients but not significance; logs change interpretation

Changing units

- $y \times c$: every coefficient and se $\times$ c; t and R^2 unchanged.
- $x_j \times c$: $\hat{\beta}_j$ and $se(\hat{\beta}_j) \div c$; t and R^2 unchanged.

Standardized (beta) coefficients

$\hat{b}_j = (\hat{\sigma}_j / \hat{\sigma}_y) \cdot \hat{\beta}_j$

One-sd change in $x_j \rightarrow \hat{b}_j$ sd change in y.

When to use logs

- % / elasticity reading; slopes invariant to rescaling; tames outliers; helps normality and homoskedasticity.
- Do not log years, percentage points, or zero/negative values; logs are hard to undo for prediction.

Model	y-side	x-side	Interpretation of β_1
Level – level	y	x	$\Delta y = \beta_1 \Delta x$
Log – level	log(y)	x	$\%\Delta y \approx (100\beta_1)\Delta x$
Level – log	y	log(x)	$\Delta y \approx (\beta_1/100)\%\Delta x$
Log – log	log(y)	log(x)	$\%\Delta y \approx \beta_1 \%\Delta x$ (elasticity)

FROM LECTURE · ECONOMETRICS 2 — LINEAR RELATIONSHIPS

Subsidy has a linear relationship with log(price), not with price.

log(sunlight) has a linear relationship with log(price); sunlight does not with price.

Battery does not have a linear relationship with log(price).

Rescaling leaves t and R^2 unchanged; linearity is judged in the transformed variables the model uses.

Sources: Chapter 2, slides 22 – 26; Chapter 6, slides 2 – 7; Econometrics 2, slides 7 – 12 · Supplement: rescaling rules, beta coefficients, level – log row

Quadratics, Interactions, Average Partial Effects

When the partial effect of x depends on the level of x or of another regressor

Quadratic

$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2, \quad \Delta\hat{y} \approx (\hat{\beta}_1 + 2\hat{\beta}_2 x)\Delta x$

Turnaround: $x^* = |\hat{\beta}_1 / (2\hat{\beta}_2)|$

Wage – experience: the wage peaks at about **24.4 years.** Does the return turn negative after that? **Not necessarily** — check how many observations lie beyond x^* and whether the quadratic is the right shape.

Housing prices with rooms and rooms²: the turnaround falls at a **low room count where few houses lie** → the negative region is practically irrelevant.

Also possible: quadratics with logs, higher polynomials.

Interaction

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + u, \quad \partial y/\partial x_2 = \beta_2 + \beta_3 x_1$

β_2 is the effect of x_2 **when $x_1 = 0$** — often not meaningful.

Reparameterize: $\cdots + \beta_3(x_1 - \mu_1)(x_2 - \mu_2)$

Then the coefficient on x_2 is its effect **at the mean of x_1,** with a ready **standard error.**

Average partial effect (APE)

Compute each unit's partial effect at its own x's with the estimates, then **average across the sample.**

FROM LECTURE · ECONOMETRICS 2 — INTERACTION

Sunlight and battery interact with each other → battery does not have a linear relationship with log(price).

With quadratics or interactions the partial effect **varies with x;** report it at meaningful values (centering) or as an APE.

Sources: Chapter 6, slides 8 – 15; Econometrics 2, slides 13 – 15 · Supplement: centered interaction form

Choosing Regressors and Prediction

R^2 is not the goal; predicting the mean differs from predicting one outcome

Adjusted R^2

$\bar{R}^2 = 1 - (1 - R^2)(n - 1)/(n - k - 1)$

Penalizes extra regressors; adding one raises $\bar{R}^2$ **iff its** $|t| > 1$. Use for nonnested models; **never compare R^2 across different y's.**

Controlling for too many factors

Do not control for the **channel** of the effect: beer consumption (fatalities on beer taxes), doctor visits (health spending on pesticides).

Adding regressors to cut σ^2

A regressor **uncorrelated with the others** lowers σ^2 without multicollinearity (beer price elasticity).

Mean prediction vs. prediction interval

Regress y on $(x_j - c_j)$: **intercept = $\hat{y}^0$ with se($\hat{y}^0$).**

$se(\hat{e}^0) = \sqrt{[\, se(\hat{y}^0)^2 + \hat{\sigma}^2 \,]}, \quad \hat{y}^0 \pm c \cdot se(\hat{e}^0)$

The PI is **wider**: $\hat{\sigma}^2$ **does not shrink as n grows.**

Predicting y when log(y) is the dependent variable

$\hat{y} = \hat{\alpha}_0 \cdot \exp(\log(y)\text{-hat}), \quad \hat{\alpha}_0 = n^{-1} \sum \exp(\hat{u}_i)$

Needs u independent of the x's; exp(log(y)−hat) alone **underestimates** $E(y|x)$. $\hat{\alpha}_0$ is **consistent, not unbiased.**

Residual analysis

Large negative $\hat{u}_i \rightarrow$ unit **below its predicted value** (e.g., an under-priced house).

Use $\bar{R}^2$ only with the **same y**; control for what the **question** requires; a prediction interval adds $\hat{\sigma}^2$.

Sources: Chapter 6, slides 16 – 25 · Supplement: interval formulas and $\hat{\alpha}_0$ (standard textbook results)